

DIGITAL FORENSICS

FINAL INVESTIGATION REPORT

PNG Image Steganography Investigation

Case Reference No.	DFIR-2026-001
Investigator	Julian Derry
Date of Analysis	February 25, 2026
Prepared For	Public
Classification	Public

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1. Overview / Case Summary

Forensic steganographic analysis of a PNG image to identify and recover concealed data embedded within image pixel channels.

Case Information

Investigator Name	Julian Derry
Date of Analysis	February 25, 2026
Case Reference Number	DFIR-2026-001
Client / Requesting Party	HIVE CONSULT
Platform(s) Examined	KALI LINUX
Tools Used	Kali Linux, ExiftTool, Steghide, zsteg, base64(GNU Core Utilities)
Report Classification	Public

2. Objectives

Identify and extract hidden or embedded information from the provided PNG image using forensic and steganographic analysis techniques, with the goal of recovering the challenge flag.

3. Evidence

List all items of evidence received, examined, or generated during the investigation.

Evidence Inventory

Exhibit ID	Description	Serial / Asset No.	Source / Location	Acquired By
red.png	Png image		https://tinyurl.com/2zp9acat	Julian Derry

5. Forensic Analysis (Steps Taken)

Describe the methodology in the order performed, sufficient for another examiner to reproduce the process.

5.1 Acquisition

The PNG image was acquired from the provided source and preserved in its original state to maintain evidential integrity throughout the examination.

5.2 Initial Inspection

Metadata Analysis

Metadata was examined using ExifTool.

Command: `exiftool red.png`

Result: No suspicious metadata or hidden flag identified.

5.3 Steghide Examination

A steganographic extraction attempt was performed using Steghide with common password guesses.

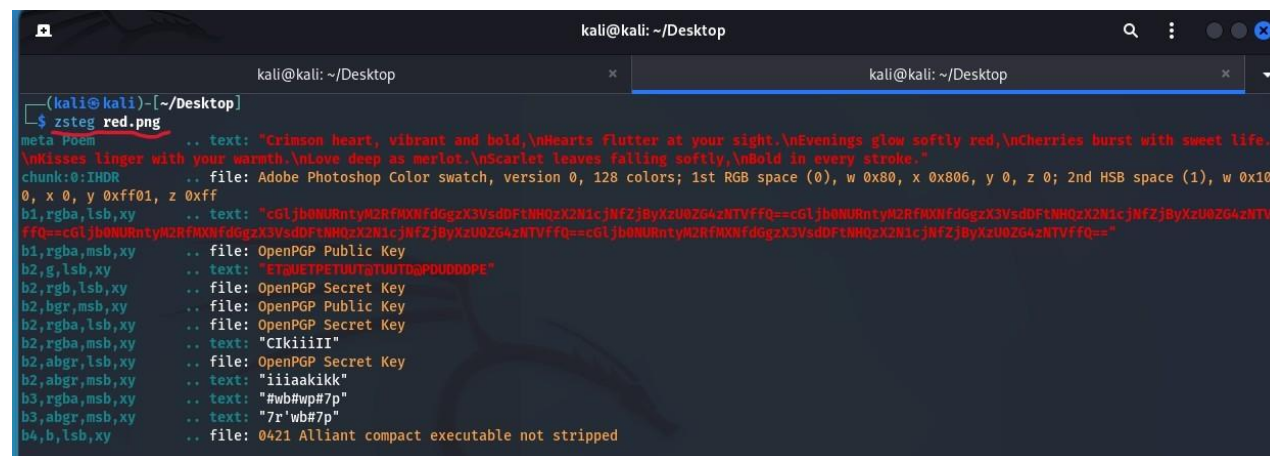
Command: `steghide extract -sf red.png`

Result: No hidden data was extracted.

Advanced Steganographic Analysis

A deep analysis was conducted using zsteg, which is designed to detect hidden data within PNG and BMP images, particularly in Least Significant Bit (LSB) channels.

Command: `zsteg red.png`



```
kali@kali: ~/Desktop
kali@kali: ~/Desktop
$ zsteg red.png
meta Poem .. text: "Crimson heart, vibrant and bold,\nHearts flutter at your sight.\nEvenings glow softly red,\nCherries burst with sweet life.\nKisses linger with your warmth.\nLove deep as merlot.\nScarlet leaves falling softly,\nBold in every stroke."
chunk:0:IHDR .. file: Adobe Photoshop Color swatch, version 0, 128 colors; 1st RGB space (0), w 0x80, x 0x806, y 0, z 0; 2nd HSB space (1), w 0x10
0, x 0, y 0xff01, z 0xff
b1,rgba,lsb,xy .. text: "cGljb0NURntyM2RfM0NfdGgzX3VsdDFtNHQzX2N1c jNfZ jByXzU0ZG4zNTVffQ==cGljb0NURntyM2RfM0NfdGgzX3VsdDFtNHQzX2N1c jNfZ jByXzU0ZG4zNTV
ffQ==cGljb0NURntyM2RfM0NfdGgzX3VsdDFtNHQzX2N1c jNfZ jByXzU0ZG4zNTVffQ==cGljb0NURntyM2RfM0NfdGgzX3VsdDFtNHQzX2N1c jNfZ jByXzU0ZG4zNTVffQ=="
b1,rgba,msb,xy .. file: OpenPGP Public Key
b2,g,lsb,xy .. text: "E1gMETPETAUT@TUTD@POUDDOPE"
b2,rgb,lsb,xy .. file: OpenPGP Secret Key
b2,bgr,msb,xy .. file: OpenPGP Public Key
b2,rgba,lsb,xy .. file: OpenPGP Secret Key
b2,rgba,msb,xy .. text: "CikiiiI"
b2,abgr,lsb,xy .. file: OpenPGP Secret Key
b2,abgr,msb,xy .. text: "iiiaakikk"
b3,rgba,msb,xy .. text: "#wb#wp#7p"
b3,abgr,msb,xy .. text: "7r#wb#7p"
b4,b,lsb,xy .. file: 0421 Alliant compact executable not stripped
```

Result: Hidden data was identified within the image's pixel LSB channels.

```

kali@kali: ~/Desktop
kali@kali: ~/Desktop
kali@kali: ~/Desktop
$ zsteg red.png
meta Poem .. text: "Crimson heart, vibrant and bold,\nHearts flutter at your sight.\nEvenings glow softly red,\nCherries burst with sweet life.\nKisses linger with your warmth.\nLove deep as merlot.\nScarlet leaves falling softly,\nBold in every stroke."
chunk:0:IHDR .. file: Adobe Photoshop Color swatch, version 0, 128 colors; 1st RGB space (0), w 0x80, x 0x806, y 0, z 0; 2nd HSB space (1), w 0x10
0, x 0, y 0xff01, z 0xff
b1,rgba,lsb,xy .. text: "cGljb0NURntyM2RfMXNfdGgzX3VsdDFtNHQzX2N1cjNfZjByXzU0ZG4zNTVffQ"
b1,rgba,msb,xy .. file: OpenPGP Public Key
b2,g,lsb,xy .. text: "ETaMETPEtU0taTUTDgPDUDODPE"
b2,rgb,lsb,xy .. file: OpenPGP Secret Key
b2,bgr,msb,xy .. file: OpenPGP Public Key
b2,rgba,lsb,xy .. file: OpenPGP Secret Key
b2,rgba,msb,xy .. text: "CkiiiiI"
b2,abgr,lsb,xy .. file: OpenPGP Secret Key
b2,abgr,msb,xy .. text: "iiiaakikk"
b3,rgba,msb,xy .. text: "#wb#wp#7p"
b3,abgr,msb,xy .. text: "7r'wb#7p"
b4,b,lsb,xy .. file: 0421 Alliant compact executable not stripped

```

6. String Decoding

The recovered encoded string was decoded using the Linux base64 utility.

Command: `echo "cGljb0NURntyM2RfMXNfdGgzX3VsdDFtNHQzX2N1cjNfZjByXzU0ZG4zNTVffQ" | base64 --decode`

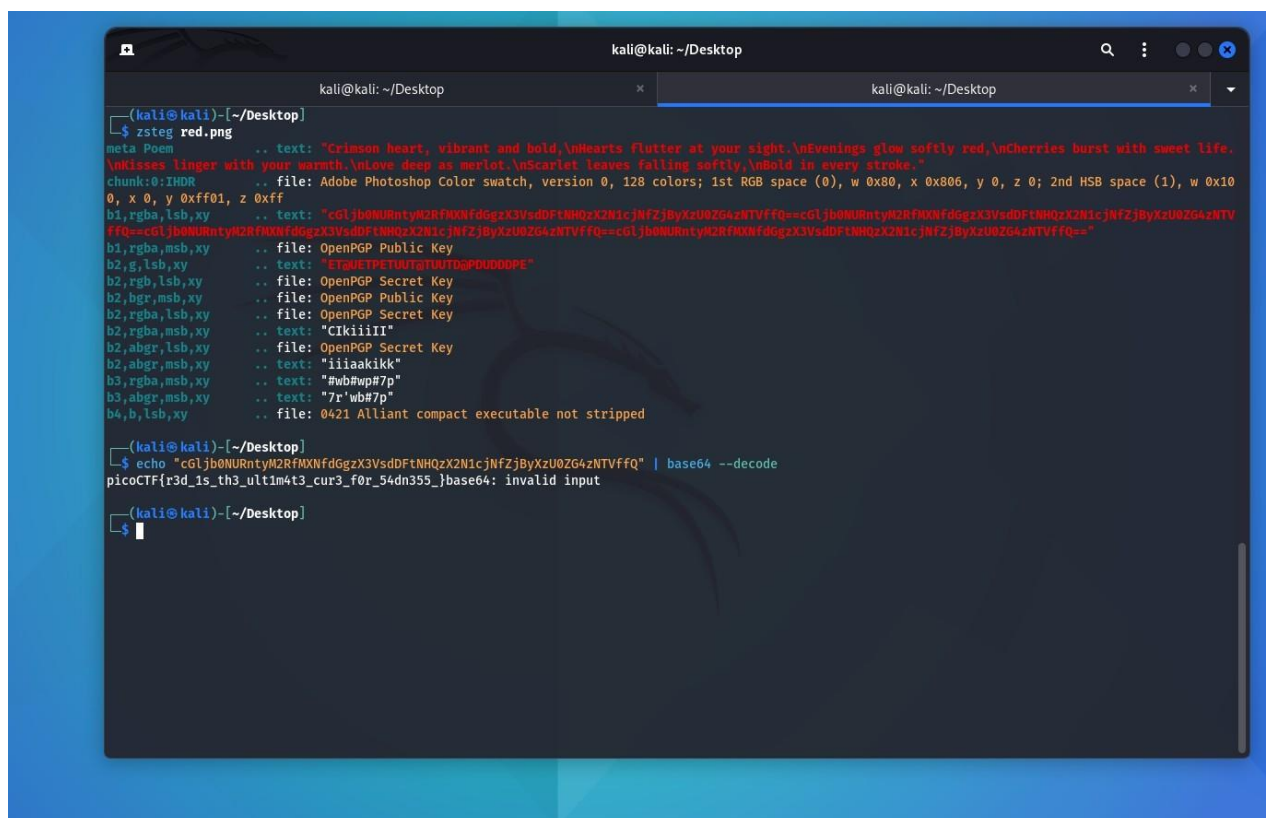
```

kali@kali: ~/Desktop
$ echo "cGljb0NURntyM2RfMXNfdGgzX3VsdDFtNHQzX2N1cjNfZjByXzU0ZG4zNTVffQ" | base64 --decode
picoCTF{r3d_1s_th3_ult1m4t3_cur3_f0r_54dn355}
kali@kali: ~/Desktop
$

```

Decoded Output

Command: `picoCTF{r3d_1s_th3_ult1m4t3_cur3_f0r_54dn355}`



7. Conclusion

The investigation successfully recovered a Base64-encoded string embedded within the image's Least Significant Bit layers. Decoding the string revealed the challenge flag:

picoCTF{r3d_1s_th3_ult1m4t3_cur3_f0r_54dn355}

The results indicate that the hidden data was embedded using Least Significant Bit (LSB) steganography, with the red color channel acting as the carrier channel for the concealed information.

8. Analyst Notes

This project demonstrates a basic forensic workflow for PNG steganography analysis:

1. Evidence preservation

2. Metadata inspection
3. Steganography screening
4. LSB analysis
5. Encoded data recovery
6. Decoding and interpretation

It is intended for educational and portfolio purposes within digital forensics and CTF context.

Report Certification

I certify that the information contained in this report is accurate and complete to the best of my professional knowledge, and that the analysis was conducted in accordance with accepted digital forensic principles.